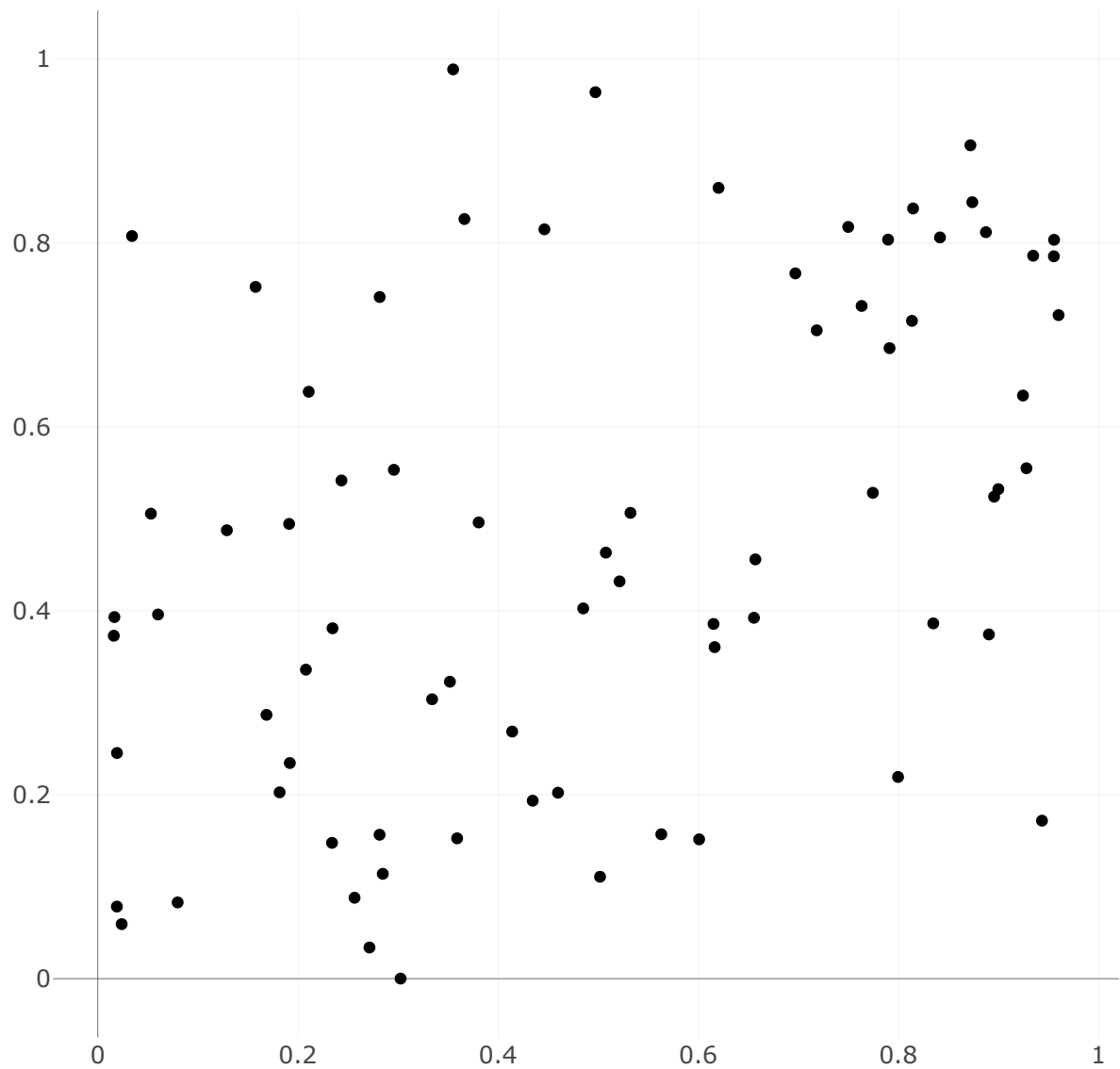


Facility Location Model

Data: demand points

The location of the demand points are randomly generated and drawn from the uniform distribution.

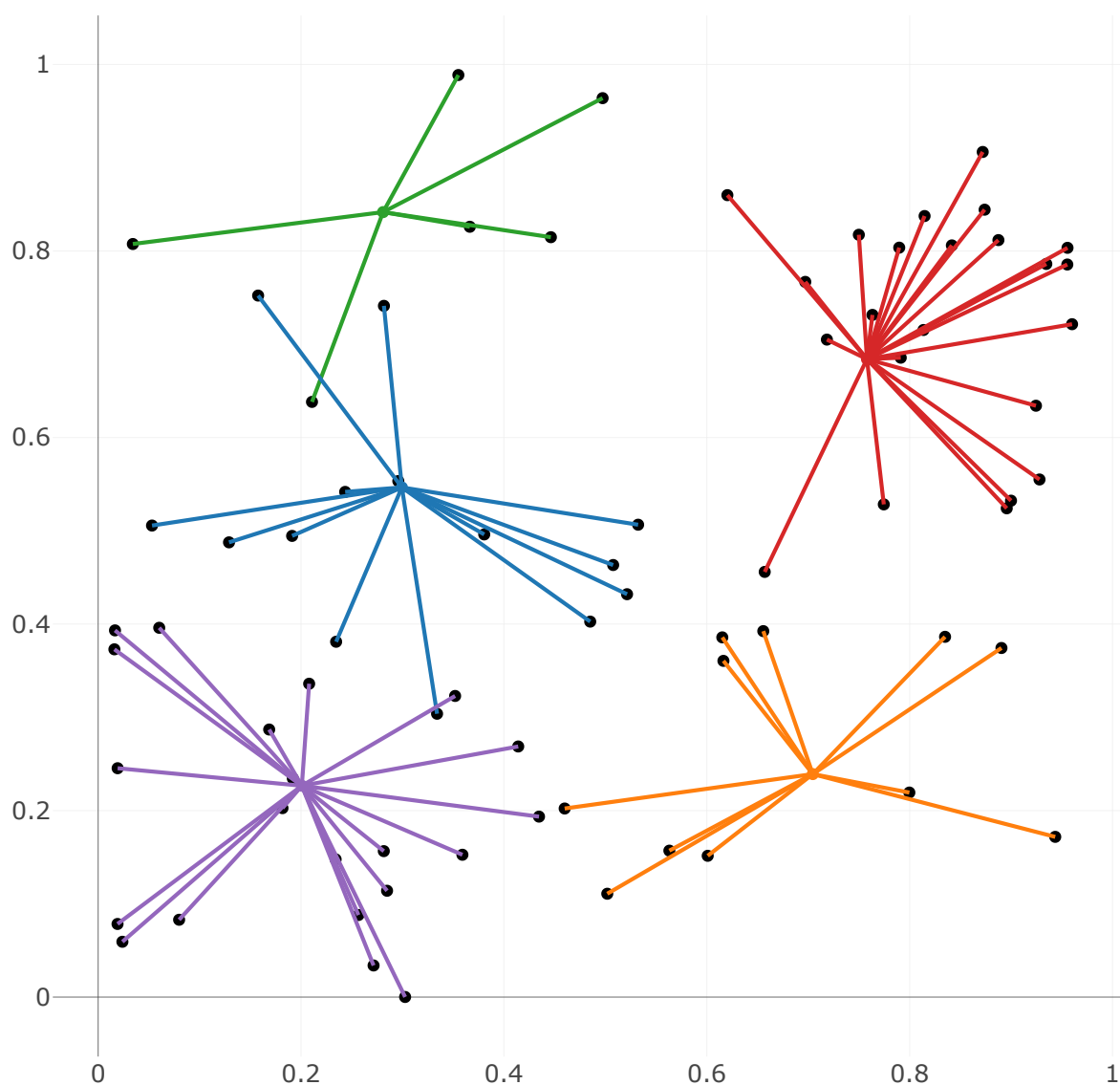
Demand points	75
Map size (width and height)	1.000
Max distance customer → facility	0.250
Max squared distance customer → facility	0.062



Model 1: results

Model 1 finds the minimum number of facilities needed to serve all customers and obey the maximum distance constraint. To make the model quadratic, the constraint is formulated as a maximum quadratic distance constraint. It is noted that the solution is **not** an assignment with shortest distances between customers and facilities.

Number facilities needed	5
Sum squared distances	2.595
Max squared distance	0.062
Solver time	11.391
Binary variables	760



Model 2: results

Model 2 find the best location of the facilities and the optimal assignment of customers to the facilities. It uses the number of facilities found in model 1.

Sum squared distances	2.236
Max squared distance	0.063
Solver time	25.610
Binary variables	375

